MIL-I-52487B(ME)
24 January 1983
SUPERSEDING
MIL-I-52487A(ME)
9 October 1975

MILITARY SPECIFICATION

INDICATORS, AIR-CLEANER, INTAKE, RESTRICTION

MECHANICAL TYPE

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers mechanical suction-pressure air-cleaner indicators for indicating operational condition of dry type air cleaners on internal combustion engines.

1.2 Classification. The indicators shall be of the following maximum signal-point capacities, as specified (see 6.2):

15 inches of water (MS53063-1)
20 inches of water (MS53063-2)
25 inches of water (MS53063-3)
27 inches of water (MS53063-4)
30 inches of water (MS53063-5)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 2940

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SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation, Methods of.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MS39158	- Adapter, Tube, Male Pipe End-Flared Type, Automotive.
MS53063	- Indicator, Air Cleaner, Intake; Restriction Mechanical Type.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The indicators shall be as shown on MS53063 and as specified herein, and shall be furnished with the fittings as specified on MS53063.

3.2 First article - preproduction. The contractor shall furnish one or more indicators of each capacity, as specified (see 6.2), for examination and testing within the time frame specified (see 6.2) to prove that his production methods and choice of design detail will produce indicators that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.5).

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3.4 Performance. The indicators shall be free of defects, shall function as specified herein, and shall not rust, corrode, or otherwise deteriorate when exposed and operated under the environmental conditions of the end item with which they are used. The indicators shall be capable of completing not less than 500,000 cycles of maximum signal travel without failure.

3.5 Case. The case shall be a completely enclosed self-contained working mechanism and shall form a tight-fitting enclosure which completely shields from corrosion, oil, dirt, moisture, air penetration and permeation, environmental deteriorating, and corrosive influences.

3.5.1 Signal window. The signal window shall be of clear shock-resistant material, free of flaws and defects, which shall retain its transparency, clarity, and original color with age. The window shall be at a minimum practical distance from the warning signal to assure freedom of moving parts and shall not break or deform under any operating condition specified herein.

3.5.2 Warning signal. The warning signal shall be bright red in color and shall be activated directly by an increase in suction pressure. The warning signal shall start to rise and be visible at not more than three-quarters of the maximum suction (maximum signal point) for which the indicator has been set, but should not be visible at less than one-half of the maximum suction; it shall rise gradually to indicate reserve life of the air cleaner, and shall lock automatically at maximum signal point and be in full view. A reset button shall allow the warning signal to return to the starting position when suction is removed.

3.5.3 Maximum signal point. The maximum point in inches of water shall be in accordance with table I.

TABLE 1. Maximum signal point.

MS Part No.	Maximum Signal Point (inches of water)	Accuracy (inches)
MS53063-1	15	± 1.5
MS53063-2	20	± 2.0
MS53063-3	25	± 2.5
MS53063-4	27	± 2.5
MS53063-5	30	± 2.5

3.6 Identification marking. Indicator shall be identified and marked in accordance with MS53063.

3.7 Vibration resistance. The indicator shall be capable of maintaining the accuracy of 3.5.3 after being subjected to the following conditions. It shall be mounted upright on a vibration table, using a fitting adapter conforming to MS39158-3, and vibrated with simple harmonic motion having an amplitude of 1/16

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inch (1/8 inch total excursion), while uniformly varying the frequency range between the limits of 10 and 40 cycles per second (CPS). The entire frequency range from 10 to 40 CPS shall be traversed, returning to 10 CPS in one minute, plus or minus 5 seconds. The indicator shall be vibrated in this manner for 24 hours in a vertical axis and 12 hours in each of the two horizontal axes. Upon completion, the indicator shall have no permanent deformation or any other damage that will affect it's operation or accuracy.

3.8 Workmanship. The indicator, including all parts and accessories, shall be constructed and finished to produce an instrument free from all defects which would affect functioning in service. Fabrication shall insure that no components or parts shall work loose or develop mechanical breakdown in normal service use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. Preproduction indicator inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 Preproduction indicator inspection.

4.3.1 Examination. The preproduction indicators shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction indicators shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling.

4.4.1.1 Examination. Sampling for examination shall be in accordance with MIL-STD-105.

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4.4.1.2 Tests. Sampling for tests shall be in accordance with MIL-STD-105, Inspection Level S-2.

4.4.2 Examination. Samples selected in accordance with 4.4.1.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for the minor defect.

4.4.3 Tests. Samples selected in accordance with 4.4.1.2 shall be tested as specified in 4.5.2.2. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The indicator shall be examined as specified herein for the following defects:

Major

101. Case not as specified.
102. Signal window not as specified.
103. Dimensions not as specified.
104. Components missing or not as specified.
105. Workmanship not as specified.

Minor

201. Identification marking missing, incomplete, or not legible.

4.5.2 Tests.

4.5.2.1 Test conditions. Ambient temperature during tests shall be 80° F, plus or minus 10° F. The indicators shall be temperature stabilized for one hour prior to testing.

4.5.2.2 Accuracy. Connect the indicator, mounted upright, to a controlled vacuum source which produces any vacuum up to 36 inches of water with an accuracy of plus or minus 0.1 inch of water. Examine for leakage in the system. Gradually increase the vacuum until the signal begins to move, then record the vacuum at that point. The vacuum must be increased slowly enough so that the inertia of the indicating system or manometer will not affect the results. Continue to gradually increase the vacuum until the signal locks up, then record the corresponding vacuum. Allow the system to return to atmospheric pressure, then push the reset button. The signal shall return to its original position. This test shall be repeated, except that the indicator shall be mounted horizontally for one series, for a total of two series. Nonconformance to 3.5.2 or 3.5.3 shall constitute failure of this test.

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4.5.2.3 Vibration. Mount the indicator upright on a vibration table supported by its pipe thread only. Subject the indicator to a simple harmonic motion having an amplitude of 1/16 inch (1/8 inch total excursion). Uniformly vary the frequency range between the limits of 10 and 40 cycles per second (cps). Traverse the entire frequency range from 10 to 40 cps and return to 10 cps in 1 minute, plus or minus 5 seconds- Vibrate the indicator for 24 hours in a vertical axis and 12 hours in each of the two horizontal axes. Upon completion, retest the indicator as specified in 4.5.2.2. Nonconformance to 3.7 shall constitute failure of this test.

4.5.2.4 Cycling. Test the indicator for accuracy, mounted upright (see 4.5.2.2), and record results, then prevent signal from making full travel (Do not block down reset button; use of a spacer is permissible.), thus preventing it from locking up. Connect indicator, mounted upright, to a vacuum source and vary the vacuum between 5 inches of water or less and 40-45 inches of water at a frequency slow enough to permit maximum travel of signal. Continue the test until 500,000 cycles have occurred, then release the signal allowing it to achieve full travel and lockup. Upon completion, retest the indicator as specified in 4.5.2.2. Nonconformance to 3.4. or 3.5.3 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

106.	Materials, methods, and containers not as specified for level A or B. Each incorrect material, method, or container shall constitute one defect.
107.	Indicator of unlike description consolidated together for level A.
108.	Quantities consolidated together not as specified for level A.
109.	Indicators of unlike description packed together for level A or B.
110.	Quantities packed together exceed the specified weight for level A.
111.	Quantities packed together exceed the gross weight or size limitation of the box for level B.
112.	Preservation and packing not in accordance with the referenced document as specified for commercial.
113.	Marking missing, illegible, incorrect or incomplete for level A, B, or commercial.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Each indicator, with connector if supplied, shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3.

5.1.1.1 Consolidation. Indicators of like description, preserved as specified in 5.1.1, shall be consolidated together in quantities of 12, in a close-fitting box conforming to PPP-B-636, W6c, style optional. Box closure and sealing shall be as specified for method V in accordance with the appendix to the box specification.

5.1.2 Commercial. Each indicator, with connector if supplied, shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Indicators of like description, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601, Overseas Type, style optional, in quantities not to exceed 250 pounds gross weight. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. Indicators of like description shall be packed together as specified in 5.2.1 except the box shall be domestic type. As an alternate, boxes conforming to PPP-B-636, V3c, may be used-when the quantity to be packed does not exceed the gross weight or size limitations of the box. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. Indicators, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188,

6. NOTES

6.1 Intended use. The indicators are intended for use with dry-type air cleaners on internal combustion engines for automotive and industrial applications.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
c. Maximum signal point required (see 1.2).
d. Time frame required for submission of preproduction indicator, and number of indicators required (see 3.2).
e. Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 Preproduction indicator. Any changes or deviations of production indicators from the approved preproduction indicator during production will be subject to the approval of the contracting officer. Approval of the pre-production indicator will not relieve the contractor of his obligation to furnish indicators conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each indicator.

6.5 Recycled material. Unless otherwise specified herein all equipment, material, and articles incorporated in the products covered by this specification shall be new, and shall be fabricated using materials produced from recycled materials to the maximum extent practicable without jeopardizing the intended use. The term "recycled materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically authorized (see 3.3).

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